<u>APPENDIX 1: Classification criteria for research areas with definitions and examples</u>

<u>Age and Growth/Life History</u>

Studies focusing on maximum age and size and/or age and size at reproductive maturity. These abstracts often focused on vertebral ring counts, and often but not always mention von Bertalanffy growth curves. Almost all had “age and growth” or “life history” in the title. Those that mentioned fecundity in addition to these characteristics were also categorized as “Reproductive Biology” (N=6). Those that also focused on the diet were also classified as Diet (N=2). Those that also focused on the Biogeography, Physiology, Population, or Population Genetics were also classified under these categories (N=1 each). Some Age and Growth/Life History abstracts mentioned Fisheries Management but not with an equivalent focus, and were therefore only classified under Age and Growth/Life History.

EXAMPLE Age and Growth/Life History abstract. “AGE AND GROWTH ESTIMATES OF THE TIGER SHARK, *GALEOCERDO CUVIERI,* AND THE SPINNER SHARK, *CARCHARHINUS BREVIPINNA*, FROM THE NORTHWESTERN GULF OF MEXICO. *Branstetter, S.*

Age and growth estimates are presented for the tiger shark, *Galeocerdo cuvieri*, and the spinner shark, *Carcharhinus brevipinna*. Young-of-the-year to large adults were examined for each species. Ages were estimated from rings or bands found in vertebral centra. Ring counts were made from the face of the centra, and compared to counts in rings in longitudinal sections. One summer opaque band and a more translucent winter band are formed each year. Verification and validation of ring periodicity are discussed. For both species, a von Bertalanaffy growth curve fit the observed data. For the tiger shark, a computer generated (Fabens, 1965) curve estimated the growth parameters as Lmax=388 cm TL, K=. 1838, to=-I. 13; a Ford/Walford plot gave a more realistic estimate of Lmax=478 cm TL. Von Bertalanffy parameters were estimated for the spinner shark using Beverton's (1954) methods: Lmax=230 cm TL, K=0.212, to= -1.94. The largest tiger shark examined (340 cm TL) was 8+ yr old, and the largest spinner shark (208 cm TL) was 10+ yr old.”

EXAMPLE Abstract categorized as both Age and Growth/Life History and Reproductive Biology: “SEXUAL MATURITY AND FECUNDITY IN THE COMMON THRESHER SHARK (*Alopias vulpinus*) FROM SOUTHERN CALIFORNIA WATERS. *Bedford, D. W.*

Common thresher sharks *(Alopias vulpinis*) 99 o, 108 +), including 18 pregnant females, were examined from southern California waters from October 1980 to November 1982. Through analysis of reproductive tissues, it was determined that males mature at a total length (TL) of 330 cm and females mature close to 390 cm TL. Pregnant females typically were found to carry four embryos. Gestation lasts approximately 9 months beginning around October. Pups are born between April and June. At birth pups are approximately 150 cm (TL). The presence of yolkcontaining egg capsules in the uteri and the discovery of large quantities of yolk and fragments of the capsules in the stomachs of embryos supply the first direct evidence of oophagy in this species. Differences in size at maturity and the number of embryos carried between common thresher sharks caught off southern California and those reported in the literature from the equatorial central Pacific and Indian Oceans suggests separate stocks."

Behavior/Behavioral Ecology

Talks that focused on the behavior of elasmobranchs other than large-scale migration or small-scale habitat usage (which were instead classified under "Movement"), mating behavior (which were instead classified under "Reproductive Biology", or prey capture (which instead were classified under "Diet/Feeding Ecology"). This category included but was not limited to swimming behavior and defensive behavior.

EXAMPLE Behavior/Behavioral Ecology abstract: "FIELD STUDIES ON THE BEHAVIOR OF THE PACIFIC ELECTRIC RAY,*Torpedo californica*: A PRELIMINARY REPORT. *Lowe*, Christopher G., Bray, Richard N., And Nelson, Donald R.*

The predatory, defensive, and space-related behaviors of Pacific electric rays are being investigated at Abalone Cove, Palos Verdes Peninsula, California. These rays appear to be the most abundant large predator in the cove. However, length of residency remains uncertain; of over 25 rays tagged at the site, none have yet been reobserved on subsequent days. During the day rays are inactive, buried in the soft sand bottom, while at night they become active and swim in the water column. Predatory and defensive behaviors are induced by presenting rays with prey items (which they shock and engulf) or by rigorously prodding them with an electrode probe. Electric discharges are recorded in situ using a housed storage oscilloscope. Recordings show differences in wave-form and pulse rates between predatory and defensive electric discharges, with some exceeding 45 volts. Acoustic telemetry, using transmitters ingested in bait, is being used to monitor diel and seasonal movement patterns."

Bioenergetics and Metabolism

Talks focused on energy gain as a result of food consumption in elasmobranchs, post capture and ingestion of prey. Bioenergetics and metabolism talks occasionally mentioned the specific diet in question and the physiology of digestion, but focused much more on Bioenergetics and Metabolism than on Diet or Physiology, and therefore were only classified as Bioenergetics and Metabolism. Most, but not all, mentioned the word "Bioenergetics" in the title.

EXAMPLE Bioenergetics and Metabolism abstract: "ESTIMATES OF DAILY RATION AND A BIOENERGETIC MODEL FOR THE BLACKNOSE SHARK, *CARCHARHINUS ACRONOTUS. Carlson, J.K. and G. R. Parsons*

Daily ration estimates and bioenergetic models for blacknose sharks, *Carcharhinus acronotus*, were developed from laboratory experiments on metabolism and growth at 28"2.0°C. Size-specific metabolic rates were determined by closed-system respirometry and growth estimated through sharks held in captivity. Daily ration was estimated for sharks 0.5-0.8 kg, 1.5 kg, and 3.5 kg using the equation of Winberg (1956). Daily ration (%body weight/day) decreased with body size and was estimated as 1.56 for smallest sharks and 0.87 for largest sharks based on a diet of menhaden, *Brevoortia* spp. General energy budgets were constructed for each size class. No estimation of reproductive investment was made because all individuals were determined to be immature. Routine metabolism generally accounted for the largest percentage of the energy budget but differed among size classes. Metabolism increased with body size from 60% of the energy budget in small sharks (0.5-0.8 kg) to 71% for the largest sharks. In contrast, growth of somatic tissue was highest in small sharks (mean=9.9%) but was 1.5% for the larger sharks."

Biogeography/Distribution

Talks that focused on global distribution of an elasmobranch species, including their preferred habitats and range. This is distinct from studies of migration behavior (which were instead classified under "Movement,"), and studies of habitat usage within an ecosystem (which were instead classified as "Behavior" and/or "Movement"). These talks occasionally but not always included discussions of systematics and taxonomy, but had much less focus on taxonomy than on biogeography, and therefore were only classified under Biogeography and Distribution not Taxonomy. Those that also focused on the Age and Growth/Life History or the Diet of the species were also classified under those categories (N=1 each). Biogeography/Distribution talks almost always included the word "Biogeography" or "Distribution" in the title.

EXAMPLE Biogeography/Distribution abstract (also classified under Taxonomy): "SYSTEMATICS AND HISTORICAL BIOGEOGRAPHY OF NEOTROPICAL FRESHWATER STINGRAYS (POTAMOTRYGONIDAE). *Lovejoy, Nathan R.*

The neotropical freshwater ichthyofauna includes representatives of some 15 primarily marine groups. However, a lack of phylogenetic and biogeographic data leaves the historical origins of these taxa unresolved - are they the result of multiple independent riverine invasions, or was a single vicariance event responsible? To begin to answer these questions, the evolutionary history of neotropical freshwater stingrays was investigated. A phylogenetic analysis of 17 stingray taxa was undertaken using 38 morphological characters. The resultant tree indicates that amphi-American Himantura represents the sister group to potamotrygonids, based on synapomorphies of the ventral mandibular musculature and the hyomandibular/mandibular articulation. Such a topology suggests that potamotrygonids are derived from a freshwater-invading ancestor distributed along the northern coast of South America prior to the emergence of the isthmus of Panama."

Biomechanics/ Functional Morphology

Talks focusing on the study of the structure and function of various parts of elasmobranch anatomy, including but not limited to teeth and jaws, muscles and fins associated with swimming, the skeletal system, and stingray spines. Biomechanics and Functional Morphology abstracts almost always mentioned "Biomechanics" or "Functional Morphology" in the title. One abstract also focused on the Reproductive Biology of a species and was also classified under that category. Several abstracts mentioned but did not equivlanelty focus on Taxonomy, and were therefore only classified under Biomechanics and Functional Morphology not Taxonomy.

EXAMPLE Biomechanics/Functional Morphology abstract: "FUNCTIONAL MORPHOLOGY OF DORSAL FINS FROM SHARKS IN THE FAMILIES CARCHARHINIDAE AND SPHYRNIDAE. *Piermarini, P.M., Pabst, D.A., and McLellan, W.A.*

In a swimming shark, the dorsal fin is a key hydrodynamic control surface that experiences the deforming forces of bending, shear, lift, and drag. Shark dorsal fins, therefore, must be designed to resist such forces. The biomechanical functions of internal supporting structures in these fins have yet to be examined. The goals of this study were to describe the morphology and function of the internal supporting structures found in shark dorsal fins. Anterior dorsal fins from five species of sharks (3 Carcharhinidae, 2 Sphyrnidae) were removed and dissected. Dissections revealed a consistent arrangement of supportive structures. Basal elements of the fin consisted of medial cartilaginous rods (pterygiophores) which resisted bending and drag forces. The pterygiophores were surrounded laterally by segmented muscle masses. Myosepta connected the lateral surfaces of the fin, and may be involved in the resistance of both shear and lift forces. Distally, the lateral muscle masses were replaced by layers of small, stiff collagenous rods (ceratotrichia), that appear to resist bending forces. Furthest distal from the base, the fin was only composed of ceratotrichia intertwined in a connective tissue matrix. This connective tissue matrix may also be involved in resisting lift and shear forces."

Conservation

Talks about the conservation of elasmobranch species, mostly focusing on legal protections distinct from fisheries management tools. One abstract each also focused on Population and Social Science, each was classified under both categories. Four abstracts also focused on Movement and was also categorized under that category. Several abstracts mentioned, but did not equivalently focus on, Fisheries Management, Ecosystem Role, and Population genetics—each was only classified under Conservation.

EXAMPLE Conservation abstract: "DEMOGRAPHY OF WESTERN ATLANTIC SAWFISHES: IMPLICATIONS FOR CONSERVATION. *Simpfendorfer, Colin A.*

Sawfish are a group of elasmobranchs that have been heavily impacted by humans due to targeted fishing, entanglement in fishing gear, and habitat modification. Four species are currently listed on the World Conservation Unions Red List of Threatened Animals, two as critically endangered. Two species occur in the western Atlantic: the small tooth sawfish (*Pristis pectinata*) and the large tooth sawfish (*P. perotteti*). There are limited biological and population data available. However, sufficient exists to construct preliminary demographic models for both western Atlantic species. A series of models were constructed to examine the sensitivity of demographic results to uncertainties in the biological and population data. The results of these models indicate that *P. perotteti* has an intrinsic rates of population increase of between 0.04 and 0.07, while *P. pectinata* has a rate between 0.08 and 0.13. These intrinsic rates of increase result in estimates of population doubling time of 10 - 17 years for *P. perotteti* and 5 - 9 years for *P. pectinata* . The implications of these results for conservation of sawfish populations are discussed."

Diet/Feeding Ecology

Talks about what elasmobranchs eat and their prey-seeking/hunting behavior. Talks were further divided into those that mention stomach content analysis, those that mention stable isotope analysis, and those that mentioned both. Two diet talks also focused on each of Age and Growth/Life History and Reproductive Biology, one also focused on each of Biogeography and Distribution, Toxicology, and Ecotourism, and four focused on Movement; these talks were classified under both. Some Bioenergetics/Metabolism abstracts, Behavioral Ecology talks, and Ecosystem Role talks mentioned but did not focus on diet, these were only classified under their primary category.

EXAMPLE Diet/Feeding Ecology abstract: FORAGING ECOLOGY OF A NEARSHORE AUSTRALIAN BATOID COMMUNITY INFERRED FROM STABLE ISOTOPIC ANALYSIS. *Jeremy J. Vaudo, Michael R. Heithaus*

Shark Bay, Australia, supports a diverse batoid fauna that show a high degree of spatial overlap. To explore whether these species are engaged in interspecific competition and how they might partition resources, we examined the foraging ecology of six batoids using stable isotope analysis. Delta 13C values indicate all species feed predominantly from a seagrass-based food web. Despite considerable overlap in delta 15N values obtained from all species, interspecific differences were detected. *Himantura fai* and *Glaucostegus typus* are slightly elevated in 15N compared to the other the species, suggesting that they feed at a higher trophic level, while *Aetobatus narinari* has lower delta 15N values compared to all other species. Examination of stomach contents suggests that similar delta 15N and 13C values are the result of similar diets. Over the size ranges examined, *H. fai, Pastinachus sephen*, and *G. typus* displayed relationships between size and both delta 15N and 13C values, suggesting ontogenetic shifts in diet. *Glaucostegus typus* tended to become less enriched in 13C and more enriched in 15N as length increased. *Himantura fai* and *P. sephen* displayed the opposite pattern with larger individuals more enriched in 13C and less enriched in 15N. With the exception of *A. narinari*, there is little evidence that Shark Bay's batoid species are partitioning food resources. Lack of resource partitioning could be the result of the high productivity of Shark Bay's seagrass beds (i.e., resources are not limiting) and population regulation occurring at life history stages not feeding on the flats we studied."

Ecosystem Role

Talks focusing on the ecological effects of the presence, decline, or recovery of elasmobranchs within an ecosystem. Some mentioned but did not focus on Diet/Feeding Ecology and Conservation, and therefore were only classified under Ecosystem Role.

EXAMPLE Ecosystem Role talk: "TROPHIC ROLE OF SHARKS IN THE ECOSYSTEM AND ITS RESPONSE TO EXPLOITATION. *García Gómez, G., and Arreguín Sánchez, F.*

Characteristics like low reproductive potentials, slow growth rates, delayed sexual maturation, as well as newborn recruitment to artisanal fisheries, they point out sharks as highly vulnerable. The objective of this study is to obtain interpretations on the role of sharks in the ecosystem, and to evaluate its answer to the exploitation, using the dynamic simulation model ECOSIM. Starting from representative solutions of 11 ECOPATH ecosystem models, ECOSIM was used to simulate increments in exploitation rate of the sharks group, evaluating the answers through changes in biomass and stability properties of the ecosystem. The groups of sharks subjected habitually to fishing, showed a higher capacity to resist increments in the exploitation rate, their responses were bigger as the fishing effort increases and finally their recovery time was low. On the other hand, strong and fast responses were presented in shark groups originally with smaller fish rate levels, the responses didn't change significantly with the increment of the exploitation rate and their recovery was slow. Contrary to the rest, in reef ecosystems the impact is important and its recovery critical, although there were not bigger changes in the ecosystem."

Ecotourism

Talks about the industry associated with human tourists paying to interact with wild elasmobranchs. One Ecotourism talk also focused on Diet and was classified under both.

EXAMPLE Ecotourism abstract: "EFFECTS OF PROVISIONING ECOTOURISM ACTIVITY ON THE BEHAVIOR OF WHITE SHARKS, *CARCHARODON CARCHARIAS. Laroche, R. Karl; Kock, Alison A.; Dill, Lawrence M.; Oosthuizen, W. Herman.*

Ecotourism operations which provide food to large predators have the potential to impact negatively on their target species, by conditioning them to associate humans with food, or by generally altering their behavioural patterns at the individual or population level. This latter effect could potentially have detrimental consequences for the predator's ecosystem, as any behavioural changes could impact on the species with which they interact. Here we present the results of a study examining the effects of provisioning ecotourism operation on the behaviour of white sharks around a small island seal colony in South Africa. Although ecotourism activity had an effect on the behaviour of some sharks (without which this ecotourism industry would not be viable), this was relatively mild, and the majority of sharks showed very little interest in the food rewards being presented. It is unlikely that conditioning would occur from the amount of ecotourism activity tested, as even those identified sharks which supplied most of the data presented here (which may possess a stronger predisposition towards conditioning, as their persistence around the boat is what allowed them to be identified) showed a nearly ubiquitous trend of decreasing response with time. Further, those sharks which succeeded in acquiring food rewards more than others demonstrated a clear ability to ignore ecotourism offerings, and typically stopped responding after several interactions. Consequently, moderate levels of ecotourism probably have only a minor impact on the behaviour of white sharks, and therefore are unlikely to create behavioural effects at the ecosystem level."

Fisheries Management

Talks about fisheries for elasmobranchs and the regulations governing those fisheries. These talks were usually but not always quite distinct from Conservation talks, and no talks were jointly classified under both (they all either focused on one topic and didn't mention the other at all, or mentioned one topic but not with an equivalent focus as the other.) Several Age and Growth/Life History and Reproductive Biology talks mentioned Fisheries Management but not with an equivalent focus, and these talks were only classified under Age and Growth/Life History or Reproductive Biology.

EXAMPLE Fisheries Management abstract: "ARTISANAL FISHERIES FOR ELASMOBRANCHS IN THE GULF OF CALIFORNIA: A MULTI-INSTITUTIONAL PROJECT. *Hueter, R. E.; Cailliet, G. M.; Márquez Farías, J. F.; Castillo Géniz, J. L.; Villavicencio Garayzar, C. J.*

Mexican fisheries rely heavily on sharks and rays and more elasmobranch tonnage is landed in the Gulf of California than in any other Mexican region. To assess the status of sharks and rays in artisanal fisheries of the Gulf of California, two U.S. institutions (Mote Marine Laboratory and Moss Landing Marine Laboratories) and two Mexican institutions (Instituto Nacional de la Pesca and Universidad Autonoma de Baja California Sur) collaborated on a two-year project in 1998-99. Seasonal surveys of the four states bordering the Gulf were conducted to characterize target species, catch, fishing effort, and other parameters. A total of 171 fishing camps with over 5,300 active boats (pangas) were found, most of which targeted elasmobranchs to one degree or another. Over the two years, 394 sampling days resulted in direct observation of 165,513 elasmobranchs of 55 species in the artisanal catch, of which 14,142 specimens (9%) were measured by researchers. Total effort, catch and CPUE Gulf-wide by season is being estimated with survey data. Results will help identify elasmobranch populations at risk and be used to formulate management strategies for conservation of sharks and rays in the Gulf of California."

Husbandry

Talks about care of captive elasmobranchs. Some Husbandry talks were also mentioned Conservation, Diet, or Reproductive Biology, but not with an equivalent focus, these were only classified under Husbandry. They were almost always given by someone whose primary affiliation was with an aquarium.

EXAMPLE Husbandry abstract: "LONG DISTANCE TRANSPORTATION OF THE SILKY SHARK, *CARCHARHINUS FALCIFORMIS. Young, F. A.; Powell , D. C.; Lerner, R.*

Pelagic silky sharks (*Carcharhinus falciformis*) are challenging to collect and transport and make an impressive display candidate for large public aquariums. Not as delicate as the scalloped hammerhead, (*Sphyrna lewini*), they do require specialized handling techniques similar to those used with *C. acronotus*. They are significantly easier to transport than *C. limbatus* of the same length. The capture, transportation and husbandry of the silky shark has been undertaken by seaside aquariums however, no long distance transportation of this species has previously been attempted. The techniques used were developed for the transportation of *S. lewini* and other small free-swimming sharks (*C. limbatus, C. acronotus and S. tiburo*). The transport equipment was designed to facilitate ram ventilation, reduce obstruction of swimming patterns and minimize the depletion of energy reserves. Transport times of 24 to 36 hours resulted in minimal risk of specimen mortality. Transport times of up to 50 hours may be possible by modifying the size and shape of the transportation vessel and reducing of the size and number of specimens transported per shipping container.

Immunology

Talks concerning medical research on elasmobranch immune systems and/or wound healing. Though the field of immunology and biomedical research in general is much broader than this, all AES abstracts about this topic fit into this narrower definition. Some talks were also classified as Physiology.

EXAMPLE Immunology abstract: "SIMILAR IN VITRO REACTIVITY OF HUMAN AND ELASMOBRANCH CELLS TO CONDITIONED MEDIA. *Grogan*, Eileen D.*

The immunology of cartilaginous vertebrates is poorly understood, and yet, research in this area may provide information pertinent to understanding the ontogenesis and phylogeny of immunity. Conditioned media generated in vitro using shark and stingray peripheral blood immunocytes are shown to be stimulatory to elasmobranch and human peripheral blood leukocytes. Reactivity suggests elasmobranchs may be used to study the evolutionary and developmental potentials of the vertebrate immune system."

Movement/Telemetry

Talks concerning the large scale (e.g., migration) or smaller scale (e.g., habitat use) of elasmobranchs. Talks were further divided into those that mention acoustic telemetry, those that mention satellite telemetry, and those that mention both. Four Movement/Telemetry talks focused equivalently on each of the Diet and Conservation categories, two focused equivalently on Population, and one each focused equivalently on Physiology and Population Genetics—each of these abstracts was classified under both relevant categories.

EXAMPLE Movement/Telemetry abstract: "MIGRATION AND HABITAT USE OF BLUE SHARKS IN THE EASTERN NORTH PACIFIC OCEAN. *Kevin Weng2, Heidi Dewar1, Suzanne Kohin1, Oscar Sosa-Nishizaki3, Erick Cristóbal Oñate González3, Barbara Block4, David Holts1, James Wraith1*

The blue shark is the most abundant pelagic shark in the world and is likely to be an important predator in open ocean ecosystems. Blue sharks have been captured in enormous numbers in high seas fisheries over many decades, sustained only by the wide distribution and high fecundity of the blue shark, but major reductions in abundance have occurred in some ocean basins and the IUCN lists the species as 'Lower Risk Near Threatened'. Knowledge of the biology of this species is important to help us understand how pelagic communities function, and why some elasmobranch species are more vulnerable to overexploitation and extinction than others. We studied the spatial ecology of blue sharks via satellite telemetry by tagging sharks in two locations within the California Current System of North America. Sharks made extensive use of the rich upwelling system off Baja California, California and Oregon, and also made long distance movements into oligotrophic regions of the subtropical gyre and the eastern tropical Pacific. Blue sharks did not appear to undertake movements in a coordinate manner as they dispersed from the tagging locations. Blue sharks inhabited cool upwelled waters, warm tropical waters to near 30°C, and waters beneath the thermocline to 7°C. The sharks undertook extensive diving to depths in excess of 700 m. Diving was frequently greatest during daytime, but sharks undertook dives at all times of the day. Knowledge of the movements and habitat usage of this abundant pelagic shark will improve our understanding of pelagic ecosystems and inform the development of high seas fishery management schemes."

Paleontology

Talks concerning the paleontology of extinct and fossil chondrichthyans. While some concerned the taxonomy of extinct elasmobranchs, here we define taxonomy as exclusively dealing with extant species, and thus talks about the taxonomy of extinct elasmobranchs were not classified as "Taxonomy."

EXAMPLE Paleontology abstract: "FRESHWATER STINGRAYS OF THE GREEN RIVER FORMATION (LATE EARLY EOCENE) OF WYOMING, WITH THE DESCRIPTION OF A NEW GENUS AND SPECIES (CHONDRICHTHYES, MYLIOBATIFORMES). *Carvalho, M. R. de, Maisey, J. G., and Grande, L.*

The late early Eocene Green River Formation (Fossil Butte Member) of Wyoming has yielded, among numerous taxa of bony fishes, two distinct forms of stingrays. One of these, *Heliobatis radians* Marsh, 1877 has been heavily sampled and is present in many private and public collections. This taxon resembles Recent dasyatids in many respects, and is known primarily from adult and sub-adult specimens of both sexes. The other stingray form is presently being described and is known from fewer specimens, both juvenile and adult of both sexes. The new genus is diagnosed on the basis of a combination of characters, including the presence of a dorsal fin anterior to the caudal stings, heavy denticulation on the dorsal surface of disc and dorsal fin, and stout tail at base. The anatomy of these taxa is described and comparisons with Recent stingray genera are conducted, along with a discussion of their phylogenetic relationships.

Parasitology

Talks about the biology of elasmobranch parasites. These were sometimes also classified as "Diet" or "Physiology" if they discussed what food items caused the parasites or any physiological effects of the parasites.

EXAMPLE Parasitology abstract: "TAPEWORMS OF ELASMOBRANCHS FROM THE GULF OF CALIFORNIA. *Jensen, K.; Caira, J. N.*

Forty-three species of sharks and rays were examined for tapeworms during the summers of 1993 and 1996.Specifically, 21 species in the superorder Galea belonging to 12 genera and 22 species in the superorder Squalea belonging to 13 genera were necropsied, their spiral intestines removed and examined for tapeworms. Of those 43 species, only 1 species (*Urobatis halleri*) had previously been examined for parasites in the Gulf of California. Twenty-one species had been examined for tapeworms elsewhere in the world. The remaining 21 species had never been examined for tapeworms. All elasmobranch species examined were parasitized by at least 1 species of tapeworm. A total of over 200 different species of tapeworms were identified, almost 50% of which are new to science, including 7 new genera. The number of different tapeworm species per species of elasmobranch ranged from 1 (*Carcharhinus porosus, C. limbatus* and *Galeocerdo cuvier*) to 16 (*Urobatis maculatus* and *U. halleri*). On average, members of the Galea were parasitized by 4 different species of tapeworms, while members of the Squalea were parasitized by 7 different species of tapeworms. Whether this tapeworm fauna is unique to the Gulf of California remains to be investigated."

Physiology

Talks about homeostasis and the biological functions of internal anatomy and biological systems of elasmobranchs. These talks include those about the physiology of blood (including stress response to fisheries capture), cells, digestive system (sometimes also mentioning but not with equivalent focus "Diet" or "Bioenergetics and Metabolism"), hormones (sometimes also classified under "Reproductive Biology" N=8) the nervous system, osmoregulatory physiology, the respiratory system, and thermal physiology. Sensory Biology and Physiology was considered distinct from this category, as was Bioenergetics and Metabolism.

EXAMPLE Physiology abstract: "INVESTIGATIONS INTO PHYSIOLOGICAL STRESS IN ELASMOBRANCHS: A HISTORICAL PERSPECTIVE. *Gregory Skomal1, John Mandelman2*

Elasmobranchs, like most fishes, are being subjected to an increasingly vast array of chronic and acute anthropogenic stressors. Although the physiological stress response in teleosts has been studied for decades, this research has lagged far behind in elasmobranchs. Of the limited number of studies conducted to date, most have centered on sharks subjected to capture and handling stress. This work has shown that sharks, like teleosts, exhibit primary and secondary responses to stress that are manifested in their blood biochemistry. The former is characterized by immediate and profound increases in circulating catecholamines and corticosteroids, which are thought to mobilize energy reserves and maintain oxygen supply and osmotic balance. Mediated by these primary responses, the secondary effects of stress in elasmobranchs include hyperglycemia, metabolic (e.g. lacticacidosis) and respiratory (hypercapnia) acidoses, and profound disturbance to ionic, osmotic, and fluid volume homeostasis. The nature and magnitude of these secondary responses are species-specific and may be tightly linked to metabolic scope and thermal physiology along with the type and duration of the stressor in question. Initial studies have also shown that the threshold to cope with, and recover from, various stressors, appears to vary interspecifically. Given the diversity of elasmobranchs, additional studies that characterize the nature, magnitude, and consequences of physiological stress over a broad spectrum of stressors are essential for the development of conservation measures. Due to K-selected life history characteristics in this group, additional studies on the sublethal impacts of various stressors on reproduction, immune function, and growth are particularly in need."

Population

Talks about assessing the population size of elasmobranchs, including increases or decreases in population over time. Studies assessing the population of elasmobranchs exclusively through population genetics methods were categorized separately as "Population Genetics," though one abstract focused on Population Genetics and other methods and was classified under both. Two Population talks each equivalently focused on Fisheries Management and Movement, and one equivalently focused on Conservation, these talks were classified under both research areas.

Some talks were also classified as "Fisheries Management" or "Conservation" if they focused on changes in population as a result of these measures.

EXAMPLE Population abstract: "ASPECTS OF THE LIFE HISTORY OF EPAULETTE SHARKS, *HEMISCYLLIUM OCELLATUM* ON HERON ISLAND REEF, GREAT BARRIER REEF, AUSTRALIA. *Heupel, Michelle R., Bennett, Michael B.*

The population of *Hemiscyllium ocellatum* on Heron Island Reef, Great Barrier Reef, Australia was examined over a three year period. The abundance of *H. ocellatum* on a 0.25 km2 site on the reef flat was examined by conducting a tag-recapture study. A total of 494 sharks were tagged during 11 sampling trips from July 1994 to August 1997. The interval between initial release and recapture of sharks ranged from 1 - 725 days. The overall recapture rate was 22% with an estimated 3% tag loss. Recaptured sharks appeared to move randomly within the study site, and distances between initial release site and recapture site ranged from 0 - 329 m. There was no obvious site attachment. Estimates of animal abundance using the Jolly-Seber method gave a range of 93 - 1359 sharks during the study period. The Peterson and Fisher-Ford methods provided ranges of 320 - 3298 and 200 - 2190 respectively."

Population Genetics

Talks using molecular ecology or population genetics techniques to study the population status or structure of elasmobranchs. These talks almost all included "Population Genetics" in their title. Two equivalently focused on Reproductive Biology, and one each equivalently focused on Age and Growth/Life History, Fisheries Management, and Movement—these abstracts were classified under both Research Areas.

EXAMPLE Population Genetics abstract: "PRELIMINARY INFORMATION ON THE POPULATION GENETICS OF THE WHALE SHARK (*RHINCODON TYPUS*). *Castro, Andrey L.F.; Stewart, Brent S.; Wilson, Steven G.; Meekan, Mark G.; Hueter, Robert E.; Bass, Anna L.; Bowen, Brian W.; Karl, Stephen*

In recent years tagging studies, including satellite transmitters, have increased our knowledge of the migratory behavior of whale sharks. These findings reinforce the perception of a mobile species that can move thousands of kilometers in short time periods. Based on the perception of high vagility, we hypothesized that inter-region or inter-ocean genetic differences for whale shark populations would be slight. Here we use mitochondrial DNA control region sequences to assess the genetic connectedness of whale sharks sampled from different oceans. We found 37 polymorphic sites (including insertions and deletions) resolving 18 haplotypes in complete mtDNA control region sequences from 30 whale sharks (10 from Gulf of Mexico; 7 from Sea of Cortez; 4 from Ningaloo Reef, Western Australia; 4 from Taiwan; 3 from South Africa; and 1 each from Maldives and Philippines). We found no significant genetic subdivision and sharing of haplotypes among ocean basins. Our data are consistent with a single global population. This inference, however, is subject to two caveats; low statistical power associated with small sample sizes, and life history considerations. First, there are marked differences observed among haplotypes, including base substitutions and gaps of 17 to 164 nucleotides, indicating that sufficient variation exists to detect population subdivision, pending larger sample sizes. Second, most samples were collected from whale shark feeding aggregations, where reproductively segregated populations could co-occur. Mixing of cohorts could contribute to the lack of phylogeographic subdivision detected here. More sequence data are being gathered to provide a statistically rigorous analysis of genetic variation among biogeographic regions, including microsatellite approaches for analysis of fine-scale breeding system."

Reproductive Biology

Studies focused on the reproduction and fecundity of elasmobranchs. Some but not all of these abstracts reference mating behavior. Some, but not all, mentioned embryonic growth. Those that also focused on life history characteristics were also categorized under Age and Growth/Life History (N=6), and several (N=8) Reproductive Biology abstracts also focused on Physiology. Three abstracts focused equivalently on Reproductive Biology and Population Genetics (N=3), two focused equivalently on Reproductive Biology and Diet, and one each focused equivalently on Reproductive Biology and Biomechanics/Functional Morphology and Population. Several Reproductive Biology abstracts mentioned Fisheries Management but not with an equivalent focus, and were therefore only classified under Reproductive Biology.

Example Reproductive Biology abstract: "THE REPRODUCTIVE BIOLOGY OF THE CHAIN DOGFISH, *SCYLIORHINUS RETIFER*. *Castro, Jose, Overstrom, Neal A., and Bubucis, Patricia.*

The chain dogfish, *Scyliorhinus retifer* is a small, comnon benthic shark of the continenta1 slopes. In spite of its abundance, its reproductive processes have not been described. Here we describe mating, egg laying, embryanic growth and development, and general biology of the species. Mating and egg layirg were observed in the laboratory. More than two hundred specimens were captured and dissected in order to study reproductive biology."

Shark Bites/ Shark Repellent

Talks that focused on negative shark-human interactions and chemical, electrical, or mechanical mechanisms to keep sharks away from humans in the water. One Shark Bites/ Shark Repellent talk jointly focused on Physiology and was classified under both.

EXAMPLE Shark Bites/Shark Repellent abstract: "ANALYTIC INDICATION OF THE IMPRACTICABILITY OF INCAPACITATING AN ATTACKING SHARK BY EXPOSURE TO WATERBORNE DRUGS: A SECOND, CONFIRMING LOOK. *Johnson, C. S., and H. D. Baldridge*

Parametric studies of three-dimensional diffusion in the ocean make it possible to estimate the release rate that would be required to maintain a desired concentration of a chemical at the outer boundary of a specified volume of water surrounding a point source under steady state conditions. For example, maintenance of a chemical (repellent) concentration of only one gram per cubic meter (i.e., approximately 1 PPM) along the outer boundary of one cubic meter of sea water would require that chemical to be released from an enclosed point source at the rate of about 800 grams per hour. Such calculations will be presented in terms of their clear implications regarding the inherent impracticability of chemical shark repellents when deployed in the "enveloping cloud" mode.”

Sensory Biology and Physiology

Talks about how the structure and function of elasmobranch sensory systems of elasmobranchs work. These are distinct from diet, behavioral ecology, biomechanics/functional morphology, and physiology talks.

EXAMPLE Sensory Biology and Physiology abstract: “VISUAL ECOLOGY OF THE SANDBAR SHARK, *CARCHARHINUS PLUMBEUS*.

Elasmobranchs rely heavily on a rich diversity of sensory input to identify prey, navigate, reproduce and interact within their environment. The role vision plays in these behaviours remains unknown for many species. This study examines the visual ecology of the sandbar shark, *Carcharhinus plumbeus*, a species which inhabits a diversity of visual environments, ranging from clear water fringing reef habitats to turbid water estuarine habitats. Here the visual ecology of the sandbar shark is assessed throughout ontogenetic development to identify visual strategies of this species in different visual environments. Using anatomical and electrophysiological techniques, key capabilities such as sensitivity to light, spatial and temporal resolution, as well as spatial and temporal summation are assessed. Topographic distributions of photoreceptor and retinal ganglion cells revealed a region of higher cell density in the centro-temporal retina, slightly ventral of the horizontal meridian. This specialisation would afford improved visual resolution in the lateral and frontal visual field with a peak spatial resolution of 5-6 cycles per degree recorded in the adult sandbar shark. Electroretinograms are used to record sensitivity to light and temporal resolution, determined by the flicker fusion frequency (FFF), at different intensities and frequencies of light stimulus. Peak light sensitivity was recorded at relatively low light intensities, indicating a high sensitivity to dim light as compared to a tuna, for instance. Peak FFF values recorded ranged between 35 to 40 Hz. Variations in temporal resolution between habitats will be discussed. This study provides insights into the visual physiology of sharks and identifies thresholds of the sandbar shark′s visual sense. This knowledge could aid our understanding of the underlying mechanisms of diurnal and nocturnal activity patterns, reproductive behaviour and predator–prey interactions in this species.”

Social Science/Human Dimensions

Talks focused on the human side of human-elasmobranch interactions, such as interviewing or surveying fishers who catch elasmobranchs, but excluding talks about sharks biting people (classified instead under “Shark Bites/Shark Repellent.” One equivalently focused on

Conservation and was classified under both research areas, some mentioned but did not equivalently focus on Fisheries Management.

EXAMPLE Social Science/Human Dimensions abstract (this one would also be classified as Conservation): "SAWFISH EXPLOITATION AND STATUS IN BANGLADESH. *Md Anwar Hossain1, Benjamin S. Thompson2, Gawsia Wahidunnesssa Chowdhury1, Samiul Mohsanin2, Zubair H. Fahad2, Heather J. Koldewey3, Md Anwarul Islam1*

Sawfish are among the world's most threatened and understudied marine fishes. There are few studies on sawfish from outside Australian and USA waters - a significant knowledge gap considering their circumtropical distribution and migratory nature. This paper presents the first assessment of sawfish exploitation and status in Bangladesh. A countrywide rapid assessment was undertaken between December 2011 and November 2012, using an interdisciplinary methodology. Fish landing stations, dry fish markets, and fishing villages were visited and a sawfish medicine maker was found and interviewed. In addition, interviews with national specialists at academic and fisheries institutions were undertaken. In total, 203 questionnaire surveys were conducted with fishers and traders in order to understand the extent of decline, potential drivers of declines, and local perceptions and uses of sawfish. Eighteen rostra were documented from museum archives and private collections, and unpublished data were sourced. Two sawfish species, *Pristis pristis* and *Anoxypristis cuspidata* were confirmed to be present in Bangladesh. General population declines were revealed. The average annual sawfish encounter rate (observations and catches) declined from 3.7 individuals using lifetime recall data (~22-year), to 1.5 using 5-year recall data, and further to 0.7 using 1-year recall data. The consensus from social research methods was that sawfish were caught as bycatch, with drift gill nets being cited as the most damaging gear type. Every respondent perceived sawfish as a useful animal. Conservation measures are proposed, including a local education and outreach programme to seek behavioural changes - primarily to release live sawfish."

Taxonomy

Talks about the taxonomic relationships between extant elasmobranch taxa and/or new species descriptions. Some Biogeography/Distribution talks mentioned but did not focus on Taxonomy.

EXAMPLE Taxonomy abstract: "THE DOGFISH FORMERLY KNOWN AS MITSUKURII: A NEW DEEP-WATER SHARK SPECIES FROM THE GULF OF MEXICO. *Mariah O. Pfleger1, R. Dean Grubbs2, Toby S. Daly-Engel1*

The shortspine spurdog shark *(Squalus mitsukurii*) is a putative circumglobal deep-water shark that was originally described from Japanese waters. Sharks of the genus Squalus are easily misidentified due to the high degree of morphological similarity with their congeners. Recent taxonomic research on this species from the Pacific has indicated that *S. mitsukurii* may actually comprise a species complex, a group of separate but closely related species. Using approximately 680 bp of mitochondrial cytochrome c oxidase 1 (barcoding gene) we found 99.61% bootstrap support for the separation of the Gulf of Mexico *Squalus cf mitsukurii* from *Squalus mitsukurii* in

Japan. We found similar support using 595 bp of mitochondrial NADH dehydrogenase 2 (ND2), as well as morphological characters. The new species in the Gulf of Mexico is named *Squalus clarkae* in honor of Dr. Eugenie Clark."

Toxicology

Talks about the effects of harmful artificial and/or natural chemicals on elasmobranchs. Some mentioned but did not equivalently focus on "Physiology," and were only classified under Toxicology.

EXAMPLE Toxicology abstract: "ORGANOCHLORINE CONTAMINANTS IN SHARKS OF THE U.S. EAST COAST. *Gelsleichter, Jim; Szabo, Nancy J.; Manire, C.A.; Morris, J.*

Even at sub-lethal concentrations of exposure, organochlorine contaminants such as pesticides and industrial chemicals pose significant health hazards to marine organisms. These compounds have been associated with a variety of health disorders in several taxa, especially those inhabiting increasingly degraded nearshore and estuarine habitats. Due to use of such regions as pupping and/or nursery grounds, certain shark species are often exposed to organochlorine contaminants at concentrations that may have detrimental effects on embryonic development, maturation, growth, and/or reproductive activity. However, despite the risks that such effects pose to these fishes, few studies have investigated the levels of these compounds in elasmobranch populations. To address such concerns, the present study describes organochlorine levels in three shark species (the bonnethead shark *Sphyrna tiburo*, blacktip shark *Carcharhinus limbatus*, and the sandbar shark *Carcharhinus plumbeus*) inhabiting estuaries and nearshore regions of the east coast of the United States. Topics including routes of exposure and potential effects of contaminant accumulation are discussed."